

FROM A DOUBLE COMPLEX TO CONVERGENCE

The purpose of this note is to explain the assertion

$$E_r^{p,q} \implies H^{p+q}(\text{Tot } K).$$

We construct the pages from a double complex and prove, under a boundedness hypothesis, that their stable terms are the associated graded spaces of total cohomology. We use the cohomological convention and the filtered-complex notation of McCleary [1, §2.2]; our K, δ_1, δ_2 denote his M, d', d'' in §2.4. All spaces are over a field $\mathbf{k}$. Throughout, the double complex is *first-quadrant*:

$$K^{p,q} = 0 \quad \text{if } p < 0 \text{ or } q < 0.$$

Indices may still be written as integers, with this zero convention. Consequently $E_r^{p,q} = 0$ whenever either index is negative. The column filtration is bounded in each total degree, as required for the convergence theorem below.

1. THE DOUBLE COMPLEX AND ITS FILTRATION

Definition 1.1 (Double cochain complex). A double cochain complex $(K^{\bullet,\bullet}, \delta_1, \delta_2)$ consists of spaces $K^{p,q}$ and maps

$$\delta_1 : K^{p,q} \longrightarrow K^{p+1,q}, \quad \delta_2 : K^{p,q} \longrightarrow K^{p,q+1}$$

satisfying

$$\delta_1^2 = \delta_2^2 = 0, \quad \delta_1 \delta_2 + \delta_2 \delta_1 = 0.$$

Its total complex $(C^\bullet, D)$ is defined by

$$C^n := \text{Tot}^n K = \bigoplus_{p+q=n} K^{p,q}, \quad D := \delta_1 + \delta_2.$$

For $n \geq 0$ this is the finite sum over $0 \leq p \leq n$; for $n < 0$, $C^n = 0$. The defining identities imply $D^2 = 0$.

Definition 1.2 (Column filtration). Define a decreasing filtration of $C^\bullet$ by

$$F^p C^n := \bigoplus_{i \geq p} K^{i, n-i}.$$

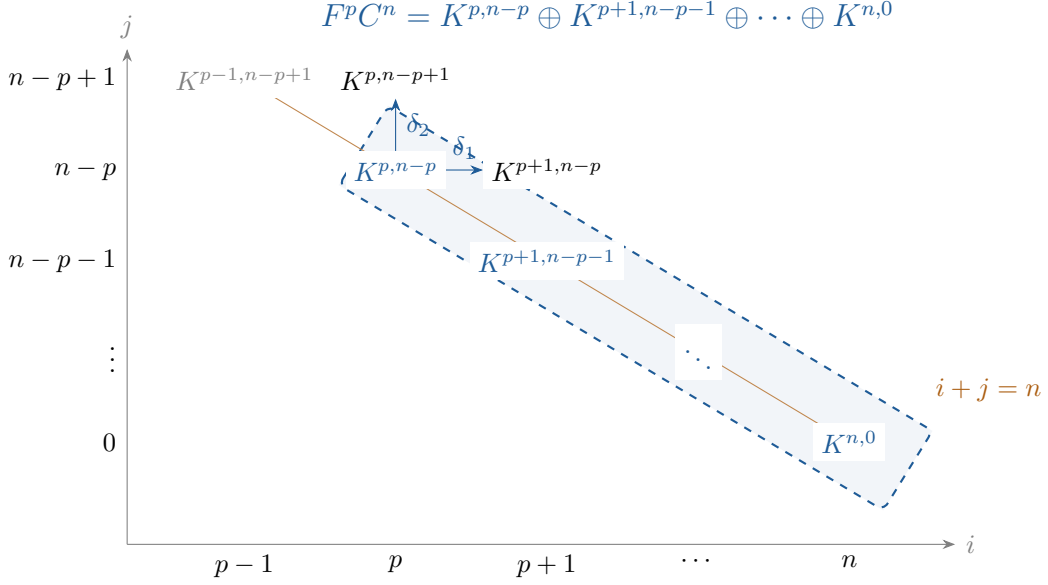
For $n \geq 0$ and $0 \leq p \leq n$, this becomes

$$F^p C^n = \bigoplus_{i=p}^n K^{i, n-i}; \quad F^p C^n = C^n \quad (p \leq 0), \quad F^p C^n = 0 \quad (p > n).$$

Thus $F^{p+1} C^n \subseteq F^p C^n$. Since δ_1 increases the column index and δ_2 preserves it,

$$D(F^p C^n) \subseteq F^p C^{n+1}.$$

In particular, each $F^p C^\bullet$ is a subcomplex.



The slanted dashed box contains the degree- n summands with $p \leq i \leq n$, ending at $K^{n,0}$.

Intermediate terms are abbreviated by dots. This is a schematic picture; when $p = n$ only $K^{n,0}$ remains, and when $p = 0$ the preceding term is zero. Both arrows increase total degree to $n + 1$ while keeping $i \geq p$; thus $D(F^p C^n) \subseteq F^p C^{n+1}$.

Definition 1.3 (Bounded filtration). A decreasing filtration on a cochain complex is *bounded in each degree* if, for every n , there are integers $a(n) \leq b(n)$ such that

$$F^{a(n)} C^n = C^n, \quad F^{b(n)} C^n = 0.$$

The bounds may depend on n . In our first-quadrant setting, $F^0 C^n = C^n$ and $F^{n+1} C^n = 0$ for $n \geq 0$; negative total degrees are zero. Thus the column filtration is bounded.

2. THE ZEROth PAGE: DEFINITION AND IDENTIFICATION

Definition 2.1 (Associated graded complex). For a filtered cochain complex $(C^\bullet, D, F)$, define

$$E_0^{p,q} := \text{Gr}_F^p C^{p+q} = \frac{F^p C^{p+q}}{F^{p+1} C^{p+q}}.$$

For the column filtration of our double complex,

$$F^p C^{p+q} = K^{p,q} \oplus F^{p+1} C^{p+q}, \quad E_0^{p,q} \cong K^{p,q}.$$

The isomorphism takes a class to its component in column p and requires no choice. We henceforth identify $E_0^{p,q}$ with $K^{p,q}$; under this identification $d_0 = \delta_2$, as proved below.

Define its differential by

$$d_0^{p,q} : E_0^{p,q} \longrightarrow E_0^{p,q+1}, \quad d_0^{p,q}(a + F^{p+1} C^{p+q}) = Da + F^{p+1} C^{p+q+1}.$$

Preservation of F makes this well-defined, and $D^2 = 0$ gives $d_0^2 = 0$.

Proposition 2.2. *For the column filtration of a double complex there is a canonical isomorphism*

$$\iota^{p,q} : K^{p,q} \xrightarrow{\sim} E_0^{p,q}, \quad a \longmapsto [a],$$

under which d_0 is identified with δ_2 .

Proof. The direct-sum decomposition gives $F^p C^{p+q} = K^{p,q} \oplus F^{p+1} C^{p+q}$. This proves the isomorphism. For $a \in K^{p,q}$, $Da = \delta_1 a + \delta_2 a$, with $\delta_1 a \in F^{p+1} C^{p+q+1}$ and $\delta_2 a \in K^{p,q+1}$. Hence $[Da] = [\delta_2 a]$, or equivalently $d_0 \iota^{p,q} = \iota^{p,q+1} \delta_2$. $\square$

Thus $E_0 = \text{Gr}_F C$ is the definition; its identification with K and the resulting description of d_0 are consequences of the column filtration.

3. THE PAGES AND THEIR DIFFERENTIALS

Definition 3.1 (Cohomological spectral sequence). A cohomological spectral sequence consists of bigraded spaces $E_r^{p,q}$, differentials of bidegree $(r, 1-r)$,

$$d_r^{p,q} : E_r^{p,q} \longrightarrow E_r^{p+r, q-r+1}, \quad d_r^2 = 0,$$

and specified isomorphisms

$$E_{r+1}^{p,q} \cong H^{p,q}(E_r, d_r) := \frac{\ker d_r^{p,q}}{\text{im } d_r^{p-r, q+r-1}}.$$

Here we include $r = 0$. The next space is obtained by taking cohomology; the next differential requires the construction below.

We now construct these data from (C, D, F) , without a boundedness assumption. The symbols Z_r, B_r below denote subspaces of C , as in the proof of [1, Theorem 2.6, pp. 34–36].

Definition 3.2 (Filtered cycles and boundaries). For $r \geq 0$ and $n = p + q$, put

$$\begin{aligned} Z_r^{p,q} &:= F^p C^n \cap D^{-1}(F^{p+r} C^{n+1}), \\ B_r^{p,q} &:= F^p C^n \cap D(F^{p-r} C^{n-1}). \end{aligned}$$

For $r \geq 1$, define

$$E_r^{p,q} := \frac{Z_r^{p,q}}{Z_{r-1}^{p+1, q-1} + B_{r-1}^{p,q}}, \quad \eta_r^{p,q} : Z_r^{p,q} \longrightarrow E_r^{p,q}$$

to be the quotient and its projection. We write $[a]_r := \eta_r^{p,q}(a)$.

The denominator lies in $Z_r^{p,q}$: its first summand has differential in $F^{p+r} C^{n+1}$, and its second summand consists of closed cochains. Notice that $Z_0^{p,q} = F^p C^n$ and $B_0^{p,q} = D(F^p C^{n-1})$.

Proposition 3.3 (The differential induced by D). *The rule $d_r[a]_r = [Da]_r$ is well-defined and gives the commutative square*

$$\begin{array}{ccc} Z_r^{p,q} & \xrightarrow{D} & Z_r^{p+r, q-r+1} \\ \eta_r \downarrow & & \downarrow \eta_r \\ E_r^{p,q} & \xrightarrow{d_r} & E_r^{p+r, q-r+1}. \end{array}$$

It satisfies $d_r^2 = 0$ and gives canonical isomorphisms $E_{r+1}^{p,q} \cong H^{p,q}(E_r, d_r)$.

Proof. For $a \in Z_r^{p,q}$, $Da \in F^{p+r} C^{n+1}$ and $D(Da) = 0$. Changing a by $b \in Z_{r-1}^{p+1, q-1}$ changes Da by an element of $B_{r-1}^{p+r, q-r+1}$; changing it by a boundary does not change Da . This proves well-definedness, and $d_r^2 = 0$ follows from $D^2 = 0$.

To identify the next page, consider the map

$$Z_{r+1}^{p,q} \longrightarrow H^{p,q}(E_r, d_r), \quad a \longmapsto \text{the cohomology class of } [a]_r.$$

It is surjective. Indeed, if $d_r[a]_r = 0$, then $Da = b + Dc$ with $b \in Z_{r-1}^{p+r+1, q-r}$ and $c \in F^{p+1} C^n$. Since $Dc = Da - b \in F^{p+r} C^{n+1}$, we have $c \in Z_{r-1}^{p+1, q-1}$, and $a - c \in Z_{r+1}^{p,q}$ represents $[a]_r$.

Its kernel is $Z_r^{p+1, q-1} + B_r^{p,q}$. To see this, a class from $Z_{r+1}^{p,q}$ is a d_r -boundary precisely when its representative belongs to $B_r^{p,q} + Z_{r-1}^{p+1, q-1}$. Intersecting with $Z_{r+1}^{p,q}$ replaces the

latter summand by $Z_r^{p+1,q-1}$, because every element of $B_r^{p,q}$ is closed. The quotient is exactly $E_{r+1}^{p,q}$. For $r = 0$, the same assertion is the usual cohomology of $\text{Gr}_F C$: its cycles are $Z_1^{p,q}/F^{p+1}C^n$ and its boundaries are $(B_0^{p,q} + F^{p+1}C^n)/F^{p+1}C^n$. $\square$

Corollary 3.4 (First two pages of a double complex). *There are natural identifications*

$$E_1^{p,q} \cong H^q(K^{p,\bullet}, \delta_2), \quad d_1[a] = [\delta_1 a], \quad E_2^{p,q} \cong \frac{\ker d_1^{p,q}}{\text{im } d_1^{p-1,q}}.$$

Proof. Apply Proposition 2.2 to $E_1 = H(E_0, d_0)$. A vertical cocycle $a \in K^{p,q}$ satisfies $Da = \delta_1 a$; Proposition 3.3 therefore identifies d_1 with the map induced by δ_1 . Its cohomology gives E_2 . $\square$

Remark 3.5 (Representative formula for higher differentials). An $E_r^{p,q}$ -class, $r \geq 1$, can be represented by $a = a_0 + \cdots + a_{r-1}$, where $a_i \in K^{p+i,q-i}$ and

$$\delta_2 a_0 = 0, \quad \delta_1 a_{i-1} + \delta_2 a_i = 0 \quad (1 \leq i < r).$$

These equations say exactly that $Da \in F^{p+r}C^{p+q+1}$. They give

$$Da = \delta_1 a_{r-1}, \quad d_r[a]_r = [\delta_1 a_{r-1}]_r.$$

Any components in columns $p+r$ and beyond may be removed from a representative: their sum lies in $Z_{r-1}^{p+1,q-1}$. Thus this formula computes the differential already defined on the quotient.

4. CONVERGENCE TO TOTAL COHOMOLOGY

The first-quadrant hypothesis ensures boundedness in each degree. This is where the relationship between the pages and actual closed total cochains becomes decisive.

Definition 4.1 (The induced filtration on cohomology). Set $H^n := H^n(C, D)$. The inclusions of subcomplexes define

$$F^p H^n := \text{im}(H^n(F^p C, D) \rightarrow H^n(C, D)).$$

Equivalently, $F^p H^n$ consists of classes having a closed representative in $F^p C^n$. Its associated graded space is

$$\text{Gr}_F^p H^n := F^p H^n / F^{p+1} H^n.$$

This filtration is fixed by (C, D, F) before any convergence assertion.

Proposition 4.2 (Stabilization of representatives). *Suppose F is bounded in each degree. Put*

$$Z_\infty^{p,q} := F^p C^{p+q} \cap \ker D, \quad B_\infty^{p,q} := F^p C^{p+q} \cap \text{im } D.$$

For each fixed (p, q) and all sufficiently large r ,

$$Z_r^{p,q} = Z_\infty^{p,q}, \quad B_r^{p,q} = B_\infty^{p,q}.$$

Consequently the stable page is canonically identified with

$$(1) \quad E_\infty^{p,q} := \frac{Z_\infty^{p,q}}{Z_\infty^{p+1,q-1} + B_\infty^{p,q}}.$$

Proof. Write $n = p + q$. For large r , boundedness gives $F^{p+r}C^{n+1} = 0$ and $F^{p-r}C^{n-1} = C^{n-1}$. The definitions of Z_r and B_r then become those of Z_∞ and B_∞ . Apply the same argument to both summands in the denominator of E_r . For sufficiently large r , that quotient is exactly (1), its differential is zero, and the next-page identifications preserve these representatives. $\square$

Definition 4.3 (Meaning of convergence in the bounded setting). For the filtered complex above, the notation

$$E_r^{p,q} \implies H^{p+q}(C, D)$$

asserts that the stable pages are naturally isomorphic to the successive quotients of the induced filtration on cohomology:

$$E_\infty^{p,q} \cong \mathrm{Gr}_F^p H^{p+q}(C, D).$$

This is the filtered-target meaning of convergence used in [1, Definition 2.4]. The symbol $\implies$ does not denote a linear map from $E_r^{p,q}$ to H^{p+q} .

Theorem 4.4 (Bounded convergence). *Let (C, D, F) be a filtered cochain complex whose filtration is bounded in each degree. There is a canonical isomorphism*

$$\theta^{p,q} : E_\infty^{p,q} \xrightarrow{\sim} \mathrm{Gr}_F^p H^{p+q}(C, D), \quad [a]_\infty \mapsto [a]_H + F^{p+1} H^{p+q}.$$

In particular, the spectral sequence converges to $H^(C, D)$ with its induced filtration.*

Proof. Let $n = p + q$. Define

$$Z_\infty^{p,q} \longrightarrow F^p H^n / F^{p+1} H^n, \quad a \longmapsto [a]_H + F^{p+1} H^n.$$

It is surjective by the definition of $F^p H^n$. Its kernel consists of those a for which $[a]_H$ has a closed representative $b \in F^{p+1} C^n$. This means exactly

$$a = b + Dc, \quad Db = 0, \quad b \in F^{p+1} C^n, \quad c \in C^{n-1}.$$

Since $a, b \in F^p C^n$, their difference Dc belongs to $F^p C^n \cap \mathrm{im} D = B_\infty^{p,q}$. The kernel is therefore $Z_\infty^{p+1,q-1} + B_\infty^{p,q}$. Taking the quotient proves the claimed isomorphism by (1). $\square$

Corollary 4.5 (First-quadrant double complexes). *If $K^{p,q} = 0$ for $p < 0$ or $q < 0$, then*

$$E_1^{p,q} \cong H^q(K^{p,\bullet}, \delta_2) \implies H^{p+q}(\mathrm{Tot} K, D).$$

For $n \geq 0$, the filtration on the target is finite:

$$H^n = F^0 H^n \supseteq F^1 H^n \supseteq \cdots \supseteq F^n H^n \supseteq F^{n+1} H^n = 0,$$

and $F^p H^n / F^{p+1} H^n \cong E_\infty^{p,n-p}$.

Proof. Each diagonal has columns $0, \dots, n$, so the column filtration is bounded in each degree. Apply Theorem 4.4. $\square$

5. WHAT CONVERGENCE DOES AND DOES NOT SAY

Remark 5.1 (Stabilization, collapse, and convergence). These assertions have different roles.

- (1) *Stabilization at (p, q)* means that the incoming and outgoing maps eventually vanish there. In the first quadrant this occurs for $r > \max\{p, q+1\}$: the incoming source has negative first index and the outgoing target has negative second index.
- (2) *Collapse at E_s* means $d_r = 0$ in all bidegrees for every $r \geq s$. This gives $E_s \cong E_\infty$; the single condition $d_s = 0$ is insufficient.
- (3) *Convergence to $H^*(C)$* identifies the stable terms with $\mathrm{Gr}_F H^*(C)$. Theorem 4.4 proves this identification; page stabilization alone does not identify a target.

Example 5.2 (Reading one total degree). For a first-quadrant double complex, total degree two gives

$$H^2/F^1 H^2 \cong E_\infty^{0,2}, \quad F^1 H^2/F^2 H^2 \cong E_\infty^{1,1}, \quad F^2 H^2 \cong E_\infty^{2,0}.$$

Thus the relevant entries lie on the diagonal $p + q = 2$. They specify the quotients in the short exact sequences

$$0 \longrightarrow F^{p+1} H^2 \longrightarrow F^p H^2 \longrightarrow E_\infty^{p,2-p} \longrightarrow 0 \quad (p = 0, 1, 2).$$

Over a field these sequences split as vector spaces, but a splitting requires choices. Convergence supplies no canonical direct-sum decomposition. If the spaces are finite-dimensional, it does give $\dim H^2 = \sum_{p=0}^2 \dim E_\infty^{p,2-p}$.

Example 5.3 (A nonzero second differential). Let the only nonzero terms be $K^{0,1} = \mathbf{k}b$, $K^{1,0} = \mathbf{k}c$, $K^{1,1} = \mathbf{k}u$, and $K^{2,0} = \mathbf{k}z$, with

$$\delta_1 b = u, \quad \delta_2 c = -u, \quad \delta_1 c = z.$$

All other values are zero. Taking vertical cohomology kills the middle column, leaving only $E_1^{0,1} = \mathbf{k}[b]$ and $E_1^{2,0} = \mathbf{k}[z]$. Thus $d_1 = 0$ and $E_2 = E_1$. However, $D(b + c) = z$, so

$$\begin{array}{ccc} \mathbf{k}[b] & & 0 \\ & \searrow d_2 = 1 & \\ 0 & & \mathbf{k}[z]. \end{array}$$

It follows that $E_3 = E_\infty = 0$. The total differential is

$$D : \mathbf{k}b \oplus \mathbf{k}c \longrightarrow \mathbf{k}u \oplus \mathbf{k}z, \quad [D] = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix},$$

which is invertible. Hence $H^*(C) = 0$, agreeing with convergence. This calculation shows why one must reach E_∞ before reading off $\text{Gr}_F H$: the nonzero entries of E_2 need not survive.

6. NATURALITY OF THE CONVERGENCE ISOMORPHISM

Proposition 6.1. *Suppose $T : C^\bullet \rightarrow C^\bullet$ satisfies $TD = DT$ and $T(F^p C) \subseteq F^p C$. It induces maps T_r on all pages satisfying*

$$T_r d_r = d_r T_r, \quad T_{r+1} = H(T_r).$$

If F is bounded in each degree and $\Phi := H^n(T)$, then

$$\begin{array}{ccc} E_\infty^{p,q} & \xrightarrow[\sim]{\theta^{p,q}} & \text{Gr}_F^p H^n \\ T_\infty \downarrow & & \downarrow \text{Gr}_F^p(\Phi) \\ E_\infty^{p,q} & \xrightarrow[\sim]{\theta^{p,q}} & \text{Gr}_F^p H^n, \end{array} \quad n = p + q,$$

commutes.

Proof. The assumptions imply $T(Z_r^{p,q}) \subseteq Z_r^{p,q}$ and $T(B_r^{p,q}) \subseteq B_r^{p,q}$. Thus $T_r[a]_r = [Ta]_r$ is defined, commutes with d_r , and respects the next-page identifications. Moreover Φ preserves $F^p H^n$, so

$$\text{Gr}_F^p(\Phi)([a]_H + F^{p+1} H^n) = [Ta]_H + F^{p+1} H^n.$$

Both paths in the square have precisely this value on $[a]_\infty$. $\square$

A bidegree-preserving map on K commuting with δ_1 and δ_2 satisfies these hypotheses. Naturality of the pages holds before convergence; bounded convergence additionally relates their stable action to the graded action on total cohomology.

REFERENCES

- [1] J. McCleary, *A User's Guide to Spectral Sequences*, second edition, Cambridge Studies in Advanced Mathematics, vol. 58, Cambridge University Press, 2001. Definitions 2.2–2.5 and Theorem 2.6 (pp. 29–37); double complexes and Theorem 2.15 (pp. 47–49); Chapter 3 (convergence). [Text available from the University of Rochester.](#)